

Symmetry and Groups — Lecture 2

IMAPP lecture notes

Spring 2026

1 The group S_3

The six permutations of three objects are

$$S_3 = \{e, (12), (13), (23), (123), (132)\}.$$

The multiplication table is obtained by composing permutations. Since $(12)(23) = (123)$ whereas $(23)(12) = (132)$, the group is non-Abelian.

2 The dihedral group D_3

D_3 is the symmetry group of an equilateral triangle. It contains three rotations and three reflections:

$$D_3 = \{e, r, r^2, s_1, s_2, s_3\}, \quad r^3 = e,$$

with presentation

$$D_3 = \langle r, s \mid r^3 = e, s^2 = e, srs = r^{-1} \rangle.$$

The rotations form a subgroup isomorphic to C_3 . The group is generated by one rotation and one reflection, so $\text{rank}(D_3) = 2$. In fact, $D_3 \cong S_3$.

3 Generators

A subset $S \subseteq G$ is a generating set if every group element can be written as a finite product of elements of S and their inverses. We write

$$G = \langle S \rangle.$$

For $C_3 = \{e, g, g^2\}$, one generator suffices: $C_3 = \langle g \rangle$ and its rank is one.

4 Homomorphisms

A homomorphism $f : G \rightarrow H$ preserves the composition law:

$$f(g_1 g_2) = f(g_1) f(g_2).$$

As an example, map D_3 to $C_2 = \{e, c\}$ by sending rotations to e and reflections to c :

$$f(e) = f(r) = f(r^2) = e, \quad f(s_i) = c.$$

The relation $s_i s_j = r^k$ ensures that the product rule is respected.

5 Kernel and isomorphism

The kernel measures which part of G collapses to the identity:

$$\text{Ker}(f) = \{g \in G \mid f(g) = e_H\}.$$

For the map above, $\text{Ker}(f) = \{e, r, r^2\}$.

An isomorphism is a bijective homomorphism. Isomorphic finite groups have identical multiplication structure even when their elements are represented differently. For example,

$$C_4 \cong \{1, i, -1, -i\} \cong \{\text{rotations by } 0, \pi/2, \pi, 3\pi/2\}.$$

6 Endomorphisms and automorphisms

An endomorphism is a homomorphism from a group to itself. An automorphism is a bijective endomorphism. The automorphisms of G form a group,

$$\text{Aut}(G) = \{f : G \rightarrow G \mid f \text{ is bijective and } f(g_1g_2) = f(g_1)f(g_2)\}.$$

For $(\mathbb{Z}, +)$, the only automorphisms are $n \mapsto n$ and $n \mapsto -n$, so $\text{Aut}(\mathbb{Z}) \cong C_2$.

7 Cayley's theorem

Every finite group of order n is isomorphic to a subgroup of the symmetric group S_n . Thus every finite abstract group can be represented as a permutation group.

Summary

S_3 and D_3 provide the central examples of small non-Abelian groups. Homomorphisms preserve structure, their kernels identify collapsed subgroups, and Cayley's theorem embeds every finite group into a permutation group.